Toward Quantum Error Correction at 100:1 Effective Overhead

Technical Research Paper v2.1

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Abstract

This paper evaluates a QONTOS path toward low-overhead fault tolerance for modular quantum systems. It does not claim that QONTOS has already achieved **100:1 effective physical-to-logical overhead**. Instead, it compares three overhead regimes - conservative, aggressive, and stretch - and examines the device, decoder, and architecture conditions required for each. The focus is on whether a hybrid code strategy combining practical surface-code ideas with more efficient LDPC-inspired approaches could reduce the physical cost of logical qubits enough to make large modular systems economically and operationally plausible.

Claim status: feasibility-envelope paper with a stretch target of 100:1 effective overhead

Keywords: quantum error correction, LDPC, surface code, decoder latency, fault-tolerant computing

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1. Introduction

Error-correction overhead is one of the central variables controlling whether large-scale quantum computing is merely scientifically interesting or actually buildable. For QONTOS, overhead is not a standalone coding-theory curiosity. It is the variable that couples:

- architecture scale
- facility size
- power budget
- capex
- algorithm viability

1.1 Claim Status Summary

Claim	Status
1000:1 is a conservative large-scale planning baseline	Derived from literature
300:1 is an aggressive QONTOS target	QONTOS target

100:1 is the QONTOS stretch target	Stretch target
QONTOS has already achieved 100:1	Not supported

1.2 Why This Matters

For 10,000 logical qubits:

- at 1000:1 overhead: 10,000,000 physical qubits
- at 300:1 overhead: 3,000,000 physical qubits
- at 100:1 overhead: 1,000,000 physical qubits

This is why the QEC story and the architecture story must stay synchronized.

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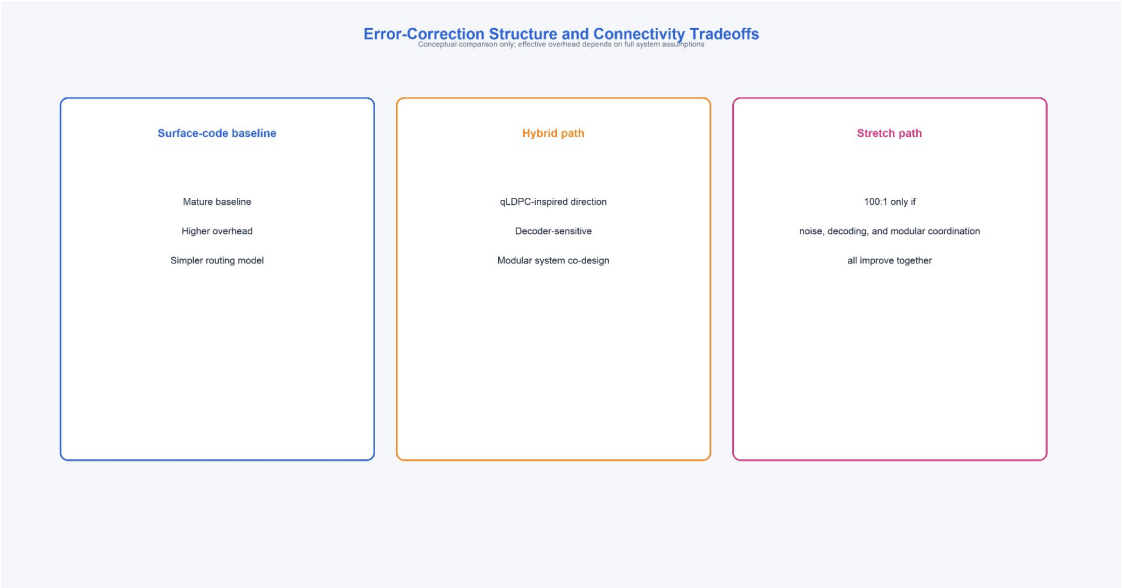
2. Overhead Scenarios



Code Structure Tradeoffs

Figure 1: Surface-code baselines, hybrid-code paths, and stretch targets should be kept conceptually distinct.

2.1 Conservative, Aggressive, and Stretch Cases

Overhead regime
1000:1
300:1
100:1

2.2 Hybrid Code Framing

QONTOS is evaluating a hybrid direction:

- surface-code-like strategies where they remain operationally practical
- LDPC-inspired or more efficient coding strategies where connectivity and decoder assumptions permit
- architecture-aware code selection for modular hardware

This should be described as a design exploration, not as a settled production code stack.

2.3 Architecture Mapping

Using the canonical stretch architecture:

Layer
Chiplet
Module
System
Data center

The paper set must keep these counts aligned everywhere.

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3. Decoder and Connectivity Requirements

3.1 What 100:1 Would Require

The stretch 100:1 case is not only a code question. It likely also requires:

- lower physical error rates than conservative baselines
- manageable 6-way or similar connectivity
- decoder latency near or below the correction-cycle budget
- tolerance to correlated noise and leakage
- modular operation without excessive communication penalties

3.2 Scenario Table

Variable
Effective overhead
Decoder latency
Decoder accuracy
Physical error
Connectivity burden

3.3 Decoder Performance Interpretation

The decoder table should not imply that `<1 us` or `500 ns` is already achieved by QONTOS. The correct framing is:

- `Base`: classical feasibility
- `Aggressive`: engineering target for modular logical qubits
- `Stretch`: required for the 100:1 architecture case

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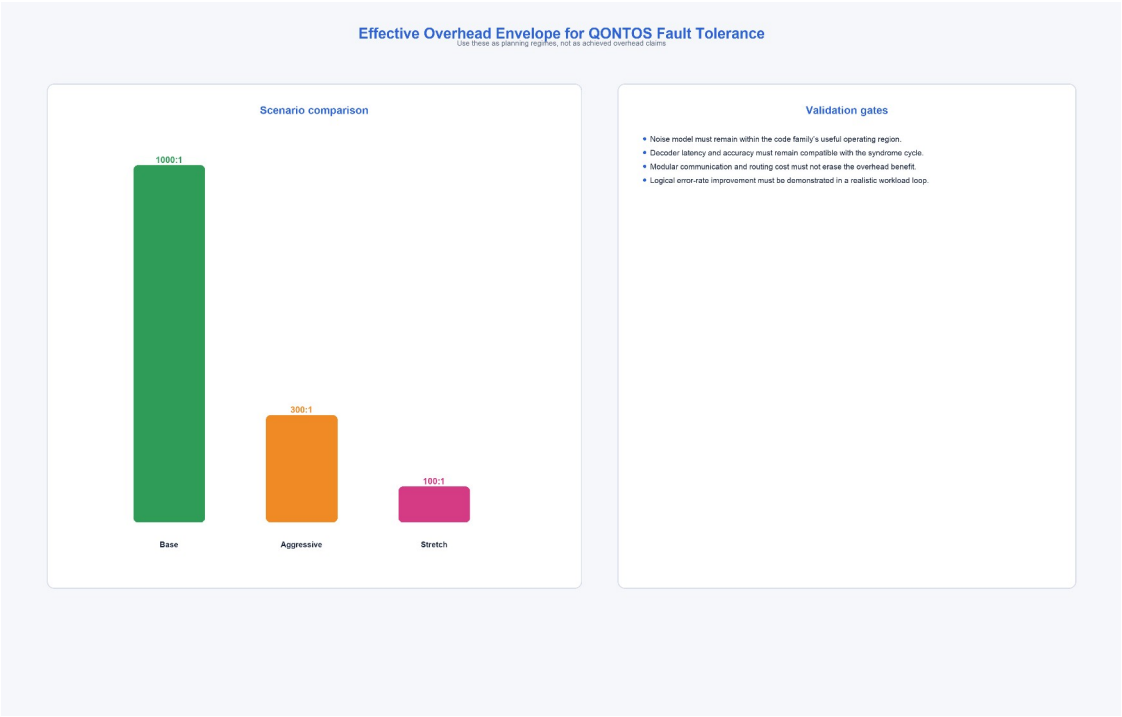
4. Implementation Strategy



Overhead Envelope

Figure 2: Effective overhead should be communicated as a planning envelope with explicit validation gates.

4.1 Syndrome Extraction

At large scale, the decoder and control path is a systems problem:

- syndrome extraction rate
- classical transport bandwidth
- local and global correction timing
- interaction with modular communication

Any low-overhead code strategy that ignores these operational costs is incomplete.

4.2 AI-Assisted Decoding

QONTOS should describe AI-assisted decoding as one possible enabler of the aggressive and stretch cases, not as a proof that those cases are already unlocked.

Recommended wording:

```
QONTOS evaluates AI-assisted decoding as a potential route to reducing
practical decoding latency and improving the viability of lower-
overhead code families in modular settings.
```

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5. Fault-Tolerance Envelope

5.1 Threshold Framing

It is reasonable to discuss threshold and margin conceptually, but this paper should avoid implying that a single threshold number settles the architecture.

Better framing:

- physical error rate and threshold margin are necessary but not sufficient
- realistic performance also depends on leakage, correlated noise, finite-size effects, and decoder behavior

5.2 Logical Error Framing

The previous draft overstated logical performance through simplified asymptotic expressions. The revised paper should say:

```
Simplified asymptotic scaling suggests that sufficiently low physical
error and sufficient code distance could drive strong logical
suppression. However, the practical logical error rate in a modular
architecture must be estimated through finite-size analysis, realistic
noise models, and decoder-in-the-loop simulation.
```

5.3 Risks and Failure Modes

Major failure modes include:

- overhead remains far above 300:1
- decoder latency dominates correction cycle time
- correlated noise invalidates simplified suppression assumptions
- modular communication overhead destroys the logical cost advantage

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6. Validation Gates

The low-overhead claim should only be promoted as a serious QONTOS program result once the following are satisfied:

1. a derivation exists for base, aggressive, and stretch overhead cases
2. decoder-in-the-loop simulation supports those cases
3. modular architecture penalties are included
4. first logical-qubit experiments align with the modeled trajectory

6.1 Phase-Aligned Gates

Phase
FOUNDATION
SPUTNIK
PIONEER
HORIZON
SUMMIT

7. Conclusion

The value of the QONTOS error-correction thesis is real, but it must be communicated precisely. The most technically defensible position today is:

- low-overhead fault tolerance is central to the architecture
- 300:1 is a meaningful aggressive target
- 100:1 remains a stretch target whose feasibility depends on device, decoder, and modular-systems performance improving together

That framing keeps the ambition while making the production plan more technically honest.

References

[1] Representative literature on surface codes, LDPC approaches, and low-overhead fault tolerance.

[2] Representative literature on decoder latency, finite-size effects, and modular fault tolerance.

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Claim posture: Feasibility envelope for aggressive and stretch QEC targets